Rank-Order Learning for MPRA Data

K562 / HepG2 / WTC11 + CAGI5 Variant-Effect Benchmark

May 2026 — Bryan Cheng

Question: do rank-aware losses beat MSE on (a) held-out lentiMPRA test sets and (b) cross-cell-type CAGI5 transfer?

Setup

Item	Value
Architecture	Prix Fixe **DREAM-RNN** (Rafi et al. 2025) — BHI + BiLSTM, ~4.2M params
Datasets	K562 (226K), HepG2 (140K), WTC11 (56K) lentiMPRA
Held-out test	Fold 0 (~10% of each)
Transfer eval	CAGI5 saturation mutagenesis (15 elements, ~14K variants)
Optimizer	AdamW, OneCycleLR, max_lr=0.005, batch=32, 80 epochs
Ensemble	9 models per loss (rotating val fold), mean prediction
17 loss variants	MSE baseline + 16 rank-loss variants

The 17 Loss Variants

Family	Variants tested	α values
MSE only	`mse` (paper baseline)	—
Plackett–Luce	`combined_pl`	0.3, 0.5
RankNet (pairwise BCE)	`combined_ranknet`	0.3, 0.5, 0.7
SoftSort (diff. sort)	`combined_softsort`	0.3, 0.5, 0.7
RankNet variants	`lambda_ranknet`, `margin_ranknet`, `sampled_ranknet`	0.3
PL variants	`weighted_pl`	0.3
Soft Spearman	`combined_spearman`	0.5
Adaptive	`adaptive_softsort`	schedule
Pure rank	`plackett_luce`, `ranknet`	—

Combined loss: `α · MSE + (1−α) · rank_loss`. Smaller α → more rank weight.

K562 Held-Out Test Set — Top 5

Method	Pearson	Spearman	Δ Pearson vs MSE
combined_spearman	**0.8245**	0.7661	+0.0012
combined_softsort_a05	0.8239	0.7656	+0.0006
combined_ranknet_a05	0.8237	0.7651	+0.0004
combined_ranknet_a03	0.8234	0.7656	+0.0001
mse (paper baseline)	**0.8233**	**0.7642**	—

Most rank losses are **within noise** of MSE on the held-out test set.
The interesting story is on CAGI5 transfer.

CAGI5 Variant-Effect Transfer — Top 7 + MSE

Method	K562-matched Sp	All-element Sp
combined_ranknet_a03	**0.4494**	0.3098
combined_softsort_a03	0.4449	0.3070
combined_lambda_ranknet	0.4430	0.3068
combined_sampled_ranknet	0.4416	0.3034
mse (baseline)	**0.4408**	0.3037
combined_pl_a03	0.4402	0.3070
`pure_ranknet`	0.4356	**0.3161** (best all-element)

+0.0086 Spearman for the best rank-loss over MSE on K562-matched elements.

CAGI5 Cross-Method Ensembling (Phase 1)

Tried multiple ensembles on the 17 trained methods:

Ensemble	K562-matched Sp
Uniform mean over all 17	0.4490
Uniform top-5 by K562 Sp	**0.4552**
Uniform top-7	0.4543
Borda rank aggregation (top 10)	0.4516
RidgeCV with K562 supervision	0.4519

Best result: uniform mean of top-5 single methods → 0.4552
(+0.0144 vs MSE baseline, +0.0058 vs best single method)

Top 5: `combined_ranknet_a03, _softsort_a03, _lambda_ranknet, _sampled_ranknet, mse`

Effect-Size Stratification

Per K562 element, bin variants by |ground truth| (quartiles).

Element	Bin 0 (small Δ)	Bin 1	Bin 2	Bin 3 (large Δ)
GP1BB	0.18	0.31	0.47	**0.61**
HBB	0.27	0.39	0.52	**0.74**
HBG1	0.21	0.34	0.49	**0.69**
PKLR	0.19	0.28	0.45	**0.57**

Large-effect variants score **0.57–0.74 Sp** — well above all-variant 0.45.
The "noise floor" on small-effect variants depresses overall metrics.

The Bets: pushing past 0.4552

After v4, three bets were attempted. **None exceeded the v4 top-5 ensemble.**
But each taught something.

Bet	Hypothesis	K562 CAGI5 Sp
v4 uniform_top5 (target to beat)	—	**0.4552 ★**
Bet 1	Heteroscedastic NLL w/ variance supervision	**0.3842**
Bet 4	RankNet weighted by large |Δy|	**0.4332**
Bet 4-rev	RankNet weighted by small |Δy|	**0.4444**

Bet 1 — Heteroscedastic NLL (FAILED)

Hypothesis: Predict per-variant aleatoric variance → downweight noisy samples → better transfer to CAGI5.

Bug found in pre-existing implementation (Feb 2026):

- `variance_prediction_r` = −0.024 to −0.032 across 3 seeds
- Variance head was learning *nothing* — root cause: `MSE(pred_var, true_var)` in linear-variance space, true_var range [0, 4.8] but most mass <0.1, so the supervision was overwhelmed by NLL

Fix: supervise in log-variance space + λ_var=2.0 (swept λ∈{2,10,50})

Result after fix:

- `variance_prediction_r` flipped: −0.029 → **+0.224** ✓
- K562 test Spearman: 0.7151 (≈ tied with broken version 0.7180)
- **CAGI5 K562 Sp: 0.3842** — worse than v4 MSE (0.4408) by **−0.057** ✗

Why it failed: `DREAM_RNN_Distributional` is a weaker architecture than Prix Fixe. The loss fix

Bet 4 — Large-Δ Weighted RankNet (FAILED)

Hypothesis: Upweight pairs with large `|relevance_diff|` → focus on dominant ranking signal that transfers to CAGI5.

Loss: `loss_per_pair = BCE × |Δy|^p` (p=1), normalized to mean 1.

Result:

- K562 test Spearman: 0.7639 (≈ tied with v4 `combined_ranknet_a03` 0.7641)
- **CAGI5 K562 Sp: 0.4332** — *worse* than v4 ranknet (0.4494) by **−0.016** ✗

Why it failed:
Variant effects are inherently **small-Δ** predictions (ref/alt differences are tiny).
Large-Δ weighting taught the model to be *less* sensitive to small differences — the opposite of what CAGI5 rewards.

→ Direction was wrong. Let's flip it.

Bet 4-reversed — Small-Δ Weighted RankNet

Hypothesis (inverted): Upweight pairs with *small* `|Δy|` → fine-grained sensitivity transfers to CAGI5.

Loss: `loss_per_pair = BCE × exp(-|Δy|/τ)` (τ=0.5), normalized.

Result:

- K562 test Spearman: 0.7624 (tied with v4)
- **CAGI5 K562 Sp: 0.4444** — **+0.011 over Bet 4** ✓ (direction confirmed!)
- But still **−0.005 vs** `combined_ranknet_a03` ✗
- And adding to v4 top-5 ensemble *hurts* (0.4552 → 0.4536) — correlated, not complementary

Conclusion: Small-Δ weighting is the right intuition, but **unweighted RankNet already implicitly learns the right pair weighting via SGD**.

Summary of K562 + CAGI5

K562 held-out test

- MSE Pearson 0.8233 / Spearman 0.7642 (paper baseline)
- All rank losses within ±0.002 of MSE on Pearson
- `combined_spearman` (Pearson) and `combined_ranknet_a03` (Sp) at the top — but the gap is in the noise

CAGI5 K562 variant effects

- MSE: 0.4408
- Best single rank loss: **0.4494** (`combined_ranknet_a03`, +0.0086)
- Best ensemble: **0.4552** (uniform top-5, +0.0144 over MSE)
- No bet exceeded 0.4552 — pair weighting tweaks ruled out

Cross-Cell-Type Generalization (NEW)

Are the K562 winners also winners on HepG2 / WTC11? Tested MSE + `combined_ranknet_a03`.

Cell type	Method	Held-out Sp	Held-out P	Δ Sp vs MSE
K562 (226K)	MSE	0.7642	0.8233	—
K562	combined_ranknet_a03	0.7656	0.8234	+0.0014
WTC11 (56K)	MSE	0.6371	0.7291	—
WTC11	**combined_ranknet_a03**	**0.6431**	**0.7339**	**+0.0060** ✓
HepG2 (140K)	MSE	**0.8010**	0.8151	—
HepG2	combined_ranknet_a03	*in progress*		

WTC11 confirms the K562 result direction. HepG2 surprisingly *higher* than K562 on MSE alone — easier task or less noisy labels?

Key Findings

1. **All rank losses are within noise of MSE on held-out Pearson** — the test set doesn't discriminate.
2. **CAGI5 transfer is where rank losses win.** Best single method: +0.009 Sp; best ensemble: +0.014.
3. **Direction of the gain transfers across cell types.** WTC11 also shows +0.006 Sp for `combined_ranknet_a03` vs MSE.
4. **Pair-weighting tweaks don't help.** Large-Δ hurts, small-Δ slightly recovers but doesn't beat unweighted.
5. **Architecture matters more than loss for CAGI5.** Bet 1's variance-head fix worked perfectly but Prix Fixe DREAM-RNN simply outperforms the distributional model.
6. **Held-out Pearson and CAGI5 K562 are anti-correlated for the top methods.** `combined_spearman` wins on Pearson but worst on CAGI5; `combined_ranknet_a03` is mid-pack on Pearson but #1 on CAGI5.

What's Next

Possible directions

- **Bet 5: AlphaGenome protocol** — deferred to "the very end" per request
- **Architecture search** — Prix Fixe DREAM-RNN may have hit its ceiling on CAGI5
- **Genuinely complementary signal** — different architectures/objectives, not loss tweaks
- **Cross-cell co-training** — train one model on K562+HepG2+WTC11 jointly

Currently running

- HepG2 `combined_ranknet_a03` (Model 5/9 in progress, ~5h to finish)
- After that: HepG2-matched CAGI5 elements (LDLR, F9, MSMB, HNF4A) eval

Reproducibility

Path	Purpose
`scripts/train_deboer_rankloss.py`	17-loss trainer
`scripts/dump_cagi5_predictions.py`	CAGI5 inference for Prix Fixe
`scripts/dump_cagi5_distributional.py`	CAGI5 inference for Bet 1 model
`scripts/ensemble_cagi5.py`	Cross-method ensemble + stratified analysis
`src/losses/ranknet.py`	Standard + large_delta + small_delta variants
`src/losses/distributional.py`	HeteroscedasticDistributionalLoss + log-var fix
`results/deboer_rankloss_1fold_v4/`	All 17 trained methods + CAGI5 predictions pkl
`results/deboer_rankloss_1fold_v5/`	Bet 4 + Bet 4-rev runs
`results/deboer_rankloss_1fold_v6_{hepg2,wtc11}/`	Cross-cell-type runs
`results/noise_resistant/DH*_v2_logvar_*`	Bet 1 (3 seeds, fixed)
`RESULTS.md`	Full results document including Bets section

Thank You

Questions?

Hyperparameters match the Rafi et al. 2025 paper (lr=0.005, 80 epochs, 10-fold) after an early bug fix where we had used lr=0.001.